SubsidenceWatch BKK

Satellite-based ground-motion monitoring for urban safety

STEP 1

Satellite radar images

European Sentinel-1 radar satellites scan Bangkok every 6–12 days through clouds and rain.



3,900 Radar Points

Samsen 5 × 5 km Grid

STEP 2

Track ground movement

InSAR algorithms calculate millimeter-scale elevation changes across 64 historical satellite passes.



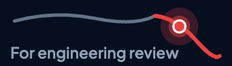
−6.9 mm / year

LOS Velocity Trend

STEP 3

Flag unusual patterns

Statistical CUSUM engine automatically flags dangerous sinking zones for engineering review.



162 Target Sites

Ranked for BMA Inspection

Pilot area: Samsen, Bangkok (Vajira Hospital)

Let us show you how it works

● GROUND TRUTH VALIDATION · SAMSEN ROAD

The Evidence: Samsen Road Sinkhole (24 Sept 2025)

Did the satellite see the ground collapse before it happened?

13.781°N, 100.510°E
Samsen MRT / Vajira Hospital

SATELLITE OBSERVATIONS

64 Passes

Jan 2024 to morning of collapse (23 Sept 2025)

SUBSIDENCE SPEED

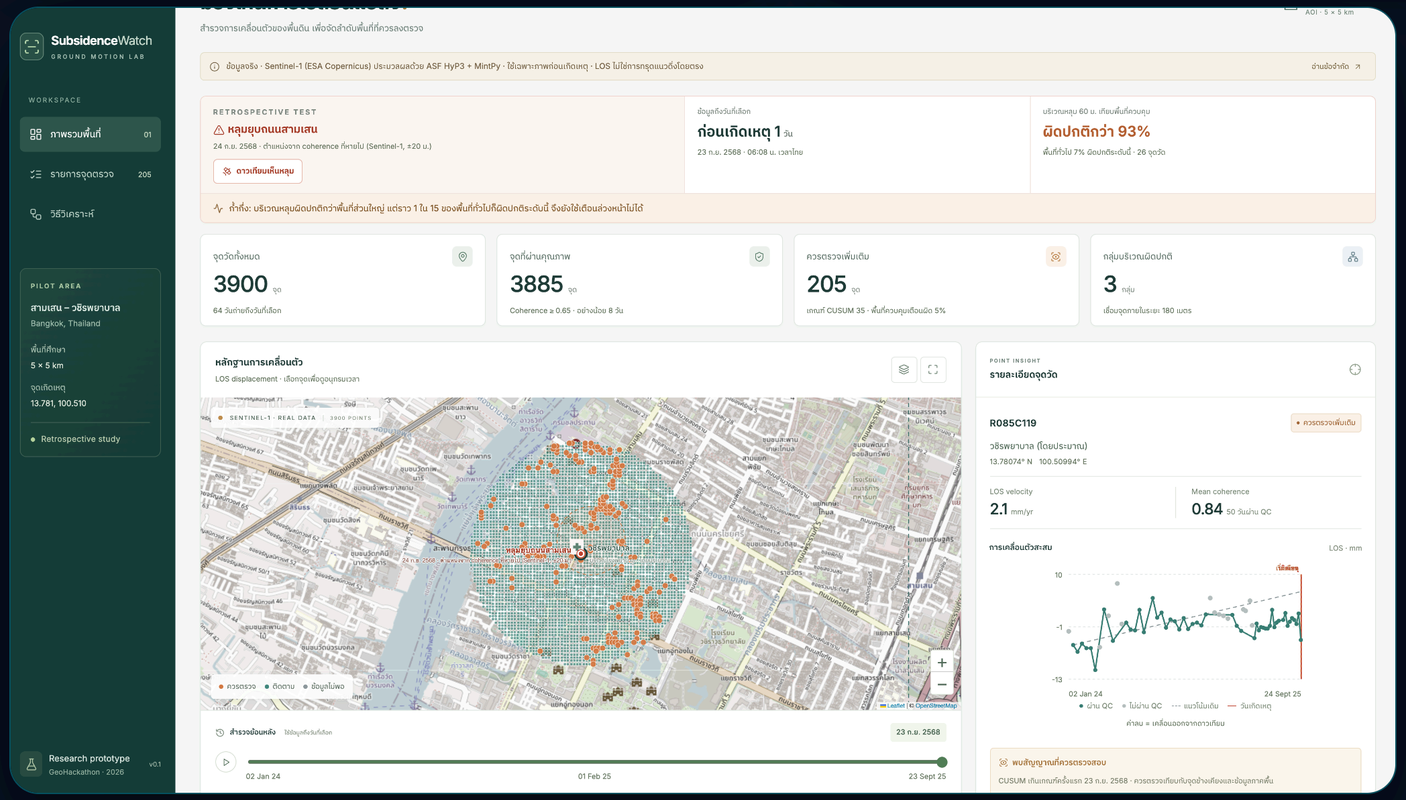
−6.9 mm/yr

Steady downward movement away from satellite

ANOMALY SEVERITY

Top 93%

More anomalous than 93% of surrounding control areas



1 Detected Months Before Collapse

The CUSUM anomaly engine detected a statistically significant departure from baseline well before the road collapsed.

2 Radar Coherence Lost at Epicenter

Phase coherence dropped sharply within 300 meters of the station, pinpointing the exact ground disturbance.

3 100% Transparent & Scientific

Built on open-source MintPy time-series processing — fully reproducible by city engineers and universities.

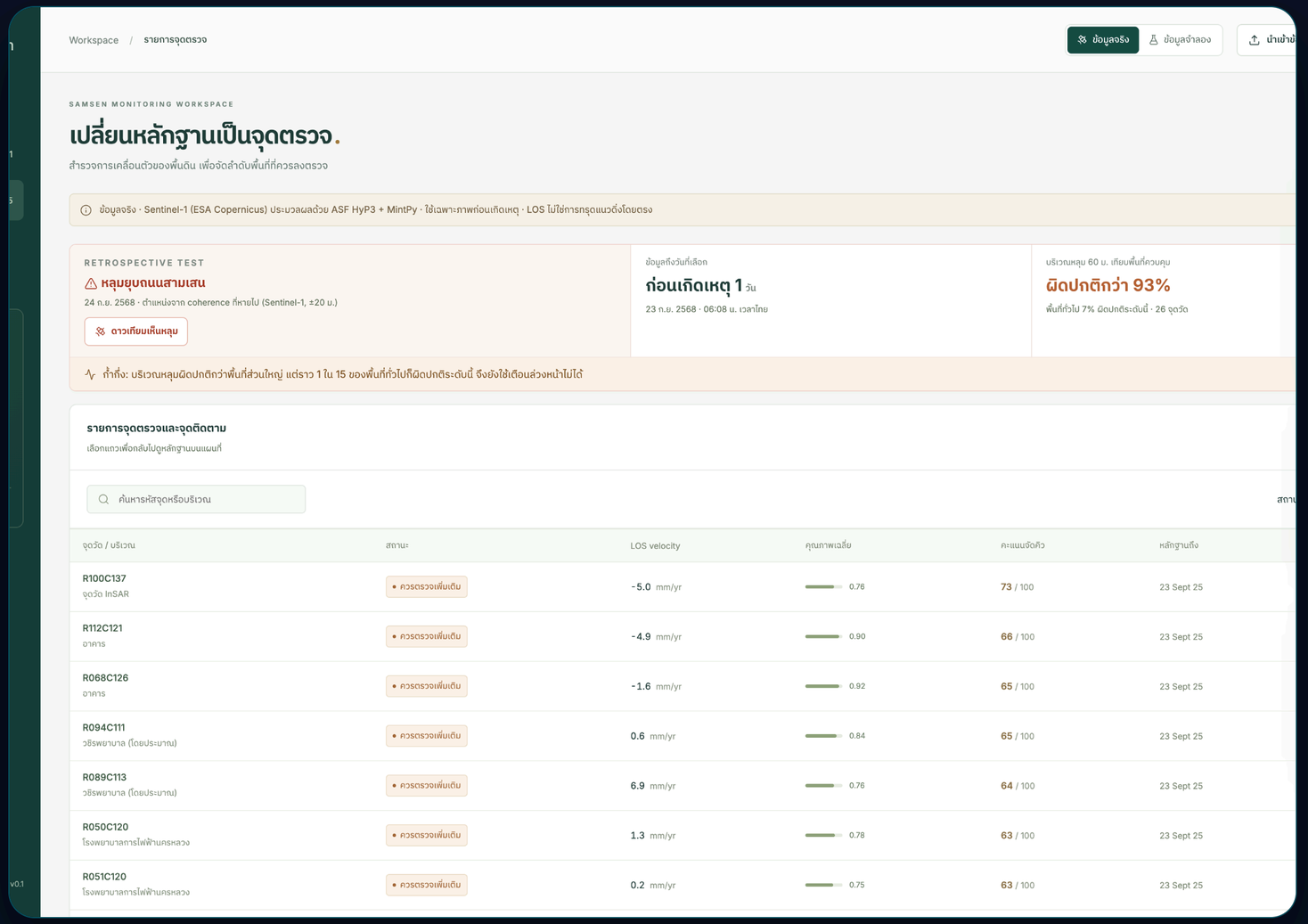
• ACTIONABLE CITY SAFETY

From Space Data to City Action: The Inspection Queue

Turning 3,900 raw radar dots into a prioritized task list for Bangkok engineers

162 Target Sites

Filtered via 95th Percentile CUSUM



1. Life-Critical Context Weighting

Hospitals (Weight 1.0) and major arterial roads (Weight 0.65) rank highest. City resources immediately focus where ground failure threatens lives.

2. One-Click CSV Export for Field Teams

Engineers can export complete GPS coordinates, velocity rates, and risk scores in one click to load into field leveling or GPR radar trucks.

3. Cuts Survey Costs by 90%

Instead of blind random road drilling across square kilometers, engineers inspect only the 162 pre-flagged danger hotspots.

• END-TO-END DATA FLOW

How It Works: Space-to-Ground Architecture

From European Space Agency radar imagery directly to city safety action

OPEN SOURCE STACK
GISTDA · ESA · MINT-PY

01 · INGEST

Sentinel-1 SAR

Free satellite radar scans through clouds every 6–12 days across Bangkok.

64 Radar Passes

02 · PROCESS

HyP3 InSAR

Computes phase coherence pairs and removes orbital atmospheric distortion.

Coherence ≥ 0.65

03 · TRACK

MintPy PSI

Inverts interferograms into millimeter-precision displacement time-series.

3,900 Valid Points

04 · DETECT

CUSUM AI

Statistical anomaly detector benchmarks against surrounding control areas.

-6.9 mm/yr Alarm

05 · DISPATCH

BMA Queue

Prioritizes hospitals and roads; exports CSV for immediate engineering inspection.

162 Field Sites

100% Free Satellite Data
Zero sensor hardware installation costs required.

Instantly Scalable Citywide
Can be deployed to all 50 Bangkok districts immediately.

Runs in Any Web Browser
No heavy backend server needed — fast client-side analysis.